



# Tagged to Stay Put

## Forecasting Missing AV Trolleys Before the Weekly Stocktake

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### Background

Shared AV trolleys migrate informally between lecture theatres, and the weekly stocktake struggles to keep pace. Little is known about what a forecasting layer does to the thing it forecasts.

### Aims

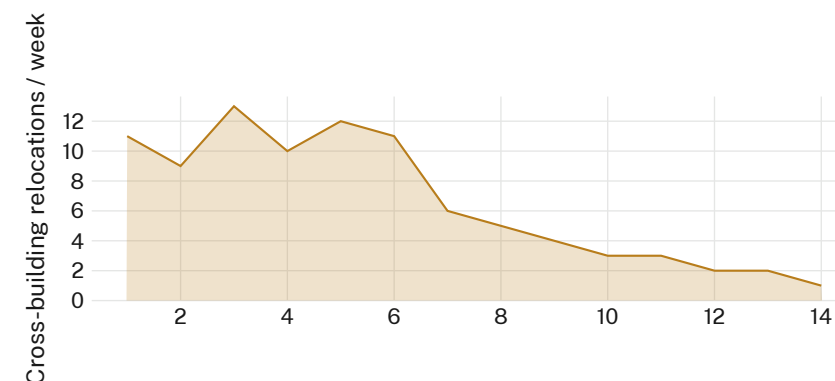
- Forecast which trolleys will be away from their bay before the stocktake
- Test whether publishing tag locations changes accuracy without changing circulation
- Establish whether “at its bay” proxies “available where booked”

### Methods

- 22 tagged AV trolleys · 3 buildings · 15m bay radius · 14-week term
- Weeks 1–6 blind logging; weeks 7–14 live last-seen map published
- Scored against the Thursday stocktake; thresholds pre-registered

### Results

Forecast accuracy rose from 71% to 94% once tag locations were published. Cross-building relocations fell by three-quarters over the same window.



Weekly cross-building relocations fell from a 6-week blind-phase mean of 11.0 to 2.75 in the eight weeks after the tracker's live map went public on 25 February 2026 (n = 22 trolleys).

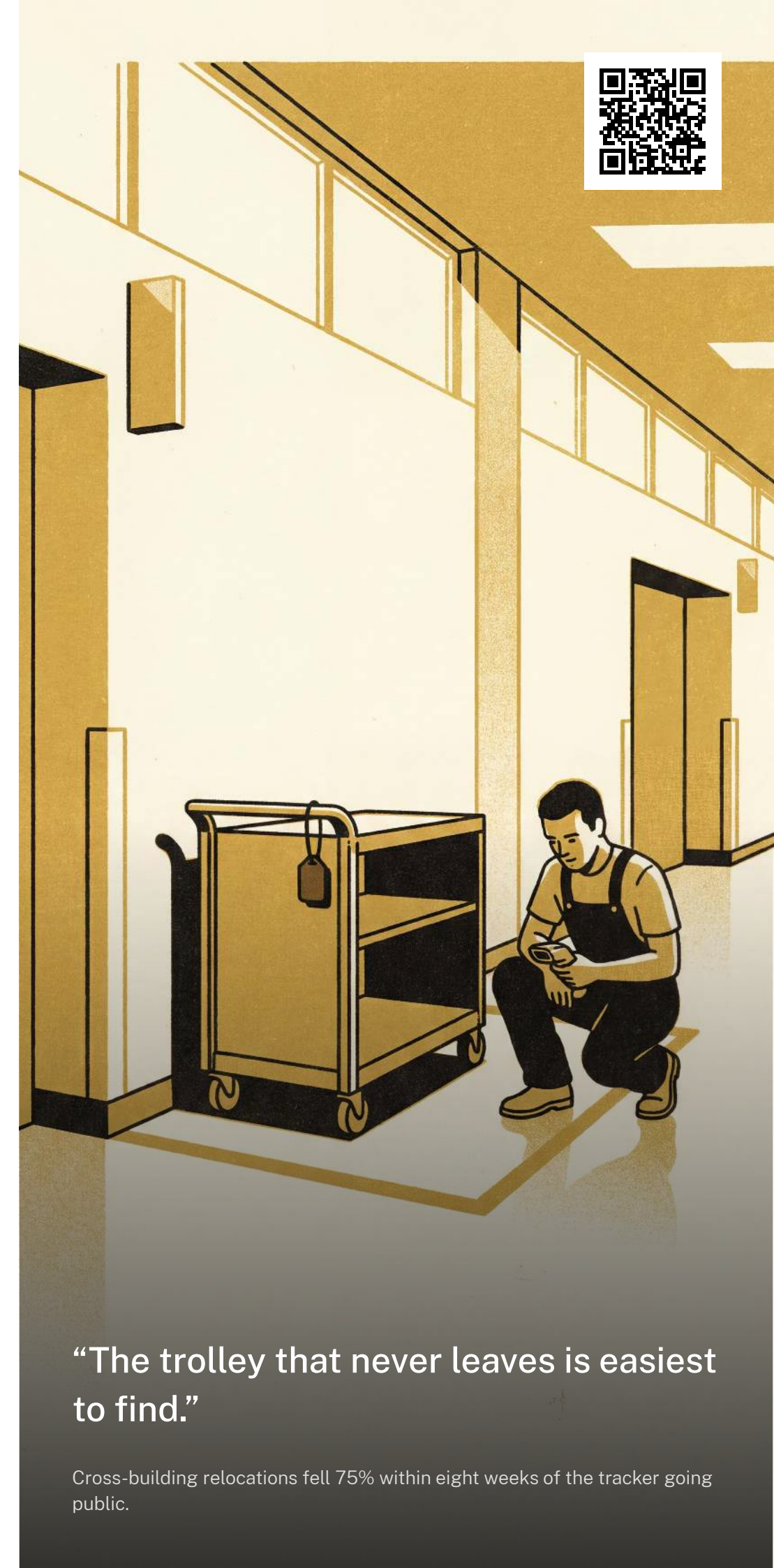
### Discussion & conclusions

The forecasting layer performs exactly as commissioned: it predicts trolley location with high fidelity. It does not predict where a trolley is needed; it predicts where one last stayed put. Next: a circulation-weighted accuracy metric, scored only on trolleys that actually moved.

### References

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- Tag data limited to trolley location; no staff movement logged. Thresholds pre-registered.

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“The trolley that never leaves is easiest to find.”

Cross-building relocations fell 75% within eight weeks of the tracker going public.